DEEP LEARNING PROJECT · NLP / SPEECH · UPV/EHU

Retrieval-Augmented Generation for Improving Noisy ASR Transcripts

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Can RAG post-correct speech recognition errors? On clean speech **yes**. On real clinical audio it can make outputs **appear semantically correct while substituting incorrect clinical entities**.

1 Motivation

Dr. García
online

3:02
"3 minutes to say yes or no?"

WHISPER-TINY OUTPUT

patient has listen oh pill for blood pressure and die a beat ease, follow up in too weeks, check haemoglobbin levels and review [inaudible] dosage

"lisinopril" transcribed as "listen oh pill" in the clinical note

ASR fails on medical vocabulary, accented speech and background noise. LLMs paraphrase instead of correcting. RAG grounds corrections in retrieved text, but **corpus choice determines whether it helps or harms.**

The core problem

BERTScore can remain high even as entity fidelity collapses. Standard metrics may **not reveal the damage** done by clinical retrieval.

2 Datasets

DATASET	SPEECH TYPE	BASELINE WER
TED-LIUM n = 10 speakers	Clean, rehearsed	0.079
MTS-Dialog n = 100 dialogues	Simulated clinical noise	0.157
PriMock57 n = 57 consultations	Real GP audio, phonetic errors	0.821

TED-LIUM and PriMock57 use Whisper-tiny ASR output. MTS-Dialog uses synthetically corrupted text (no audio).

Evaluation Metrics

WER ↓ WORD ERROR RATE

Word-level edit distance between output and reference. Can exceed 1.0 when output is longer than the reference.

(substitutions + insertions + deletions) / reference length

BERTScore ↑ SEMANTIC SIMILARITY

Cosine similarity between RoBERTa token embeddings of output and reference. Rewards fluent paraphrase.

A wrong drug name can score the same as the correct one if it fits the context.

NER F1 ↑ MEDICAL ENTITY FIDELITY

Span-level F1 over clinical entities (diseases, chemicals) extracted by scispaCy from output and reference.

Catches entity substitutions that BERTScore misses. Essential for clinical evaluation.

Project Repository
Code, evaluation scripts and experimental configurations

3 Hypotheses

- H1 ✓ RAG improves surface accuracy (WER) over LLM-only on clean speech (TED-LIUM)
- H2 ✓ Domain-specific retrieval outperforms generic retrieval on clean speech
- H3 ✓ Dense retrieval outperforms BM25 on the same corpus (Wilcoxon p = 0.002, TED-LIUM)
- H4 ✗ RAG does not improve correction on real clinical audio (PriMock57, Wilcoxon p < 0.001)

4 System Architecture · 2 Methods × 3 Corpus Tiers

Audio → Whisper-tiny (ASR) → Chunk (400w / 50w overlap) → Retrieve (top-k = 3) → Mistral 7B (Correct)

RETRIEVAL CONFIGURATIONS

Gen: Wikipedia · Rel: clinical text corpus · Mat: leave-one-out in-dataset (LOO)

BM25 LEXICAL
- C3-Lex-Gen : Wikipedia
- C3-Lex-Rel : Domain-relevant
- C3-Lex-Mat : Domain-matched (LOO)

vs

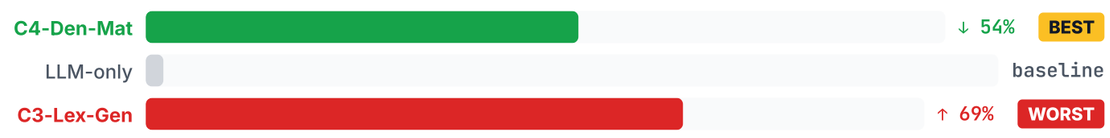
DENSE (MINILM + FAISS)
- C4-Den-Gen : Wikipedia
- C4-Den-Rel : Domain-relevant
- C4-Den-Mat : Domain-matched (LOO)

5 Results: Clean Speech · TED-LIUM (n = 10)

DENSE DOMAIN-MATCHED RAG IMPROVES SURFACE ACCURACY

↓54%	↓62%	↑69%
WER VS LLM-ONLY C4-DEN-MAT, WILCOXON P ≤ 0.05	DENSE BEATS BM25 SAME CORPUS, WILCOXON P = 0.002	WER WORSE GENERIC BM25

RELATIVE WER VS LLM-ONLY BASELINE · 3 KEY CONDITIONS SHOWN



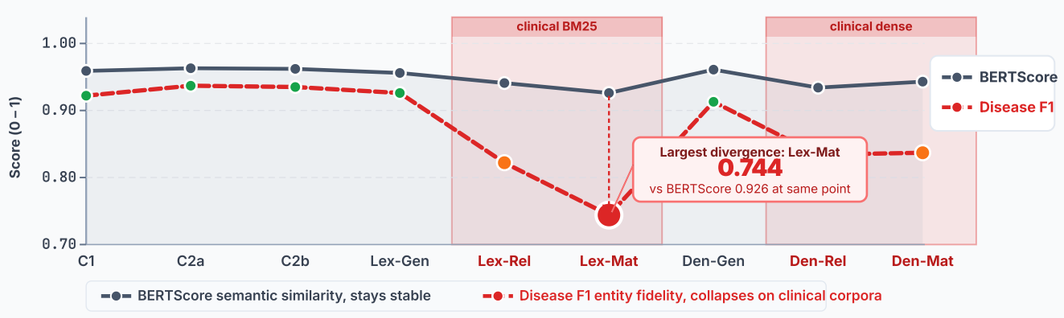
6 Key Insight: The Clinical Corpus Penalty · MTS-Dialog (n = 100)

CRITICAL FINDING: SEMANTIC SIMILARITY (BERTSCORE) STAYS HIGH. MEDICAL ENTITY FIDELITY (NER F1) COLLAPSES.

Clinical retrieval surfaces **plausible but wrong medical entities.** The LLM substitutes them into uncertain ASR tokens. BERTScore rewards contextual fluency and cannot detect the substitution.

BERTSCORE	DISEASE F1 (NER)
0.93 to 0.96	0.94 to 0.74
stable · misleadingly high	collapses · hidden damage

Semantic similarity vs medical entity fidelity · all 9 MTS-Dialog conditions

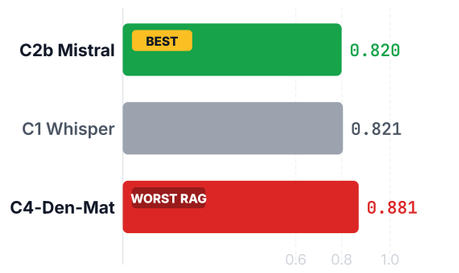


7 Results: Real Clinical Audio · PriMock57 (n = 57)

NO RAG CONDITION BEATS LLM-ONLY ON REAL CLINICAL AUDIO

↑≈7%	0
WER DEGRADATION, C4-DEN-MAT VS LLM-ONLY WILCOXON P < 0.001	RAG VARIANTS BEATING BASELINE (1 RAG variant evaluated at n=57: C4-Den-Mat)

WER · FULL RUN (N = 57) · 3 CONDITIONS REPORTED



Full 57-consultation run. Only C1, C2b and C4-Den-Mat reported at n = 57. Phonetic errors cannot be recovered from text context alone (p < 0.001).

8 RAG Failure Examples · C4-Den-Mat

REFERENCE		RAG OUTPUT
hypertension correct condition	→	**hypotension** opposite meaning
autoimmune disease general category	→	**sarcoidosis** specific wrong entity
No family history. explicit statement	→	**[fabricated epilepsy paragraph]** hallucinated from retrieved note

All invisible to BERTScore (0.93 to 0.96). All detected by Disease F1 (entity fidelity drops to 0.74). **Contextually plausible. Clinically wrong.**

9 Conclusions

- ✅ **Clean speech:** Dense domain-matched RAG reduces WER by 54% (Wilcoxon p ≤ 0.05)
- ❌ **Clinical speech:** Domain-matched retrieval actively harms medical entity fidelity
- **BERTScore is insensitive to entity identity** in clinical settings. Medical entity fidelity (NER F1) is required for evaluation.

10 Future Directions

- → **Entity-aware retrieval** Filter retrieved passages to those sharing entities with the ASR hypothesis, reducing incorrect clinical substitutions
- → **Phonetic RAG** Combine acoustic signal with text context to recover phonetically grounded errors that text-only methods cannot fix
- → **GraphRAG** Introduce structural constraints during retrieval to prevent plausible but incorrect entity substitution
- → **Larger ASR models** Whisper-medium or Whisper-large-v3 to reduce the phonetic error rate before post-correction

RAG can make clinical transcripts *look right*, yet be *clinically wrong*.

ASR: Whisper-tiny · **LLM:** Mistral 7B and LLaMA 3 8B · **Embeddings:** all-MiniLM-L6-v2 + FAISS · **Data:** TED-LIUM · MTS-Dialog · PriMock57 · **Eval:** WER · BERTScore · scispaCy Disease / Chemical F1

Claims apply to Whisper-tiny + Mistral 7B under this retrieval setup; not a general statement about RAG.